# CancerOS UI UX and Interactive Demonstration Specification

Document ID: COS-SPEC-009

Version: 1.0

Date: 22 October 2026

Status: Proposed design and implementation baseline

Scope: Breast cancer research prototype and hackathon demonstration

## 1 Purpose and product experience

CancerOS shall present a time-indexed breast cancer research case through an interactive lesion viewer, clinical charts, source records and evidence-linked explanations. The interface must let a researcher answer four questions: What is available? When was it collected? What changed? Which findings are observed, derived or simulated?

The main workspace combines one selected patient, one explicit history context and one selected snapshot. The user may inspect actual history or choose a hypothetical branch without changing the authentic baseline. A polished display must preserve source identity, physical scale, missingness and uncertainty. The product remains a research prototype, with no diagnosis, prescription, treatment ranking or validated patient-specific response claim.

This specification defines screen behavior, routes, controls, visual tokens, charts, accessibility, failure handling, API integration and a rehearsable demonstration. It is a design contract, not a claim that these features have already been built or user-tested. It follows COS-SPEC-008 where the architecture's earlier examples differ in naming, dates, quantities, state ownership or endpoints.

MUST and SHALL mark acceptance requirements. SHOULD marks the intended default. Wireframes are schematic layouts, not medical images or implemented screens. Any example case identifiers, quantities and times are illustrative; public demonstration values must come from approved persisted records.

## 2 Source review and scope decisions

| Source issue or requirement | UI decision |
|---|---|
| **Architecture uses short provenance names; database contract uses canonical names** | Display readable labels mapped explicitly in section 5; preserve canonical values in application data. |
| **A snapshot can combine an observed event and derived measurement** | Show both the history context and individual item origin. Do not assign every item the snapshot's origin. |
| **Timeline must synchronize modalities with different collection dates** | Selected snapshot time stays prominent; each modality shows its actual time and whether it is current or historical context. |
| **Four positions per hypothetical branch include a real baseline** | Show one shared observed baseline followed by at least three simulated follow-ups; baseline never becomes synthetic because a branch is selected. |
| **MRI phases and real follow-up visits are different** | DCE phase selector lives inside the imaging panel, below the examination selector. It never adds timeline visits. |
| **Source clinical data may be sparse** | Missing modalities are explicit empty states. Do not populate real charts with synthetic substitutions. |
| **Anatomy may be illustrative** | Label its layer Illustrative context; do not call it patient anatomy unless image-derived and verified. |
| **Forecasting, interpolation and automated segmentation are optional** | Omit unavailable actions from primary navigation. P0 supports precomputed masks/meshes and discrete state navigation. |
| **Branch comparison is P1 in the PRD but useful to a presentation** | P0 compares two states within a history; side-by-side branches are P1. A demo can switch branches sequentially. |

| Source issue or requirement | UI decision |
|---|---|
| **API defines metadata but lacks evaluation and capability discovery routes** | Specify explicit amendments in section 17; never invent evaluation results or imply those endpoints already exist. |
| **Research editors add observations while visitors are viewers** | Judge access defaults to read-only; live writes use an authorized presenter and an isolated demo project. |

P0 includes case selection, summary, twin workspace, chart/table views, two-state comparison, five scenario families, run and validation status, observation submission, interpretation/evidence inspection, limited import administration, and a guided demo. A compact evaluation page is required, using actual versioned reports when available. Advanced radiology workstations, mask editing, arbitrary patient chat, whole-body reconstruction and therapy recommendations are outside P0.

# 3 Users and navigation architecture

| User | Primary needs | Available actions |
|---|---|---|
| **Research viewer or judge** | Inspect approved cases, move through time, distinguish evidence types | Read published cases and branches, inspect permitted sources and explanations |
| **Research editor or presenter** | Add source-supported observations, run configured scenarios, inspect processing | Viewer actions plus approved writes, simulation and interpretation requests |
| **Research administrator** | Curate sources, review QC, publish imports, investigate failures | Editor actions plus import approval, model/source configuration and withdrawal |

Global navigation contains Cases, Guided demo and Evaluation. Editors also see Activity; administrators see Data administration. The case shell contains Overview, Twin workspace, Scenarios and Sources. Interpretations are a contextual panel in the workspace with a deep link, rather than an unrelated chatbot screen. The active case label and current history context remain visible throughout case navigation.

| UI route | Screen and behavior |
|---|---|
| **/projects/{project}/cases** | Case library; preserve safe filter state on return |
| **/projects/{project}/cases/{patient}/overview** | Research case profile |
| **/projects/{project}/cases/{patient}/twin** | Main viewer; query state, branch, run, lesion, panel and compare_state identify selection |
| **/projects/{project}/cases/{patient}/scenarios** | Branch library and generation workflow |
| **/projects/{project}/cases/{patient}/sources** | Approved source metadata and provenance |
| **/projects/{project}/demo** | Guided sequence over approved case/snapshot IDs |
| **/projects/{project}/evaluation** | Versioned evaluation report and limitations |
| **/projects/{project}/activity** | Authorized job list or current-session job references |
| **/projects/{project}/admin/imports** | Staged imports and validation/publication workflow |

These are browser routes, not REST endpoint names. Route IDs remain subject to server authorization. Sensitive text, signed asset URLs and note contents must never enter route query strings. Reload and browser Back restore the authorized case, context and snapshot. Camera rotation stays transient and is not a clinical state update. Do not silently replace a deep-linked withdrawn state with a different snapshot; explain its status and offer the latest accessible state.

# 4 Screen inventory and layout

## 4.1 Case library

The library opens with the title Research cases, a short research-only disclosure, search by display code, and filters for source, available modalities and genuine longitudinal coverage when supported by the API. Present a compact table on desktop and labeled cards on narrow screens. Fields are case code, known subtype, side, available examinations, published

synthetic branches and data availability. Unknown subtype reads Not recorded. No fabricated age, avatar, diagnosis confidence or patient photograph is shown.

Primary action is Open case. A secondary Start guided demo opens the approved sequence. Administrators can select Import data. Show actual counts returned by authorized queries; do not advertise 100 cases or 500 branches as already loaded. Empty filters show No cases match these filters and Clear filters. An unpopulated project says No approved cases yet with the correct role-specific next action.

Search and modality-count filters require server support or operate over a fully loaded bounded demo manifest; do not pretend a current page is the entire cohort. Pagination follows the contract cursor and snapshot revision. Case cards never show counts aggregated across inaccessible data.

## 4.2 Case overview

Header shows pseudonymous code, source collection/release, known breast side and a source-status badge. Summaries show pathology-supported subtype and receptor statuses with an explicit Unknown option. Separate source facts from calculated measurements. A baseline section shows source time basis, selected lesion, annotation meaning, measurement and method. Below it, show actual history availability, synthetic branch count, missing modalities and source-rights summary.

Primary action Open twin opens the selected baseline or latest observed state according to an explicit user choice. Secondary actions Explore scenarios and View sources preserve case context. A prominent Actual history available indicator is based on verified repeated observations, not simply more than one MRI phase. It does not claim clinical validation.

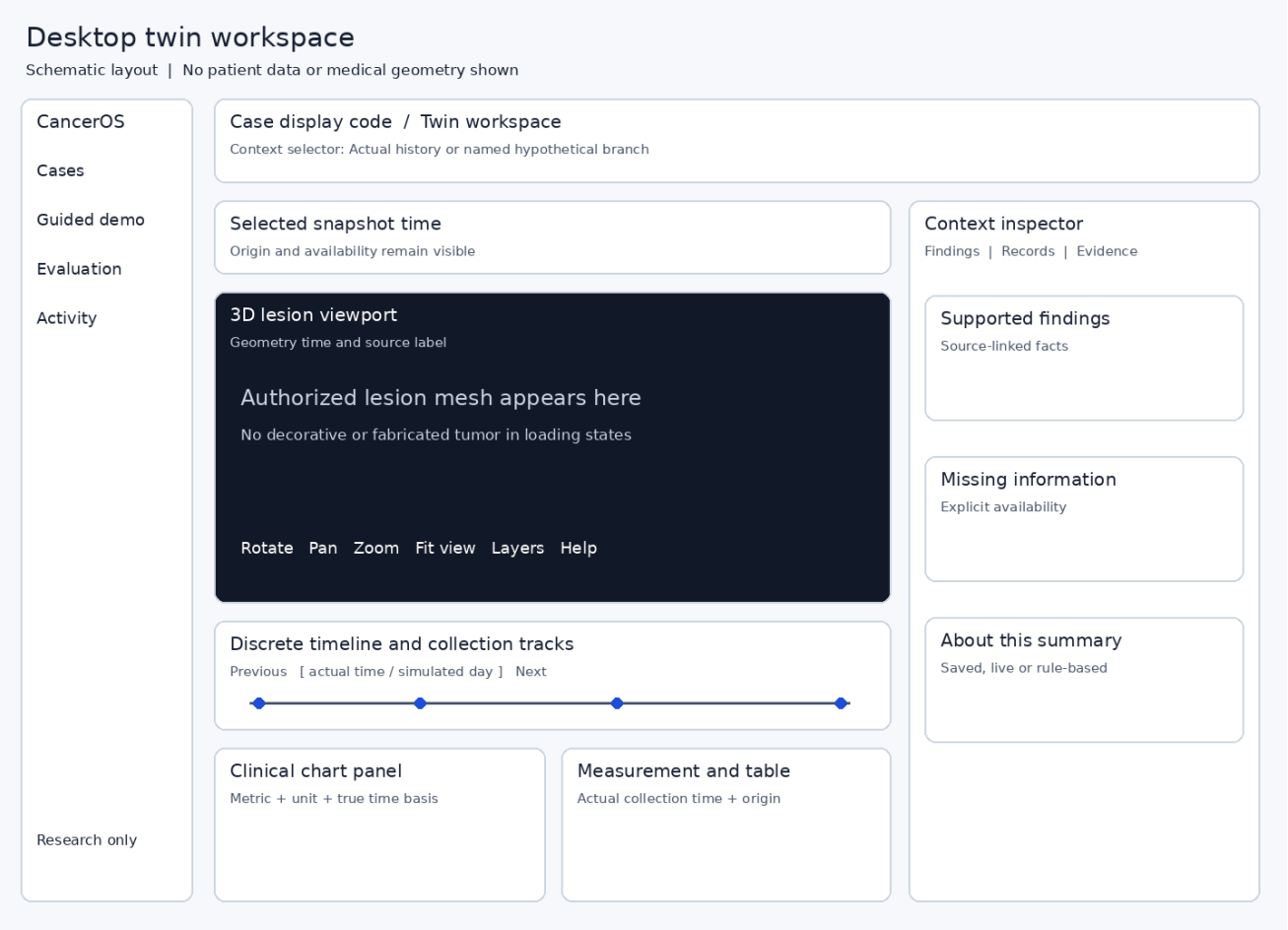
## 4.3 Twin workspace

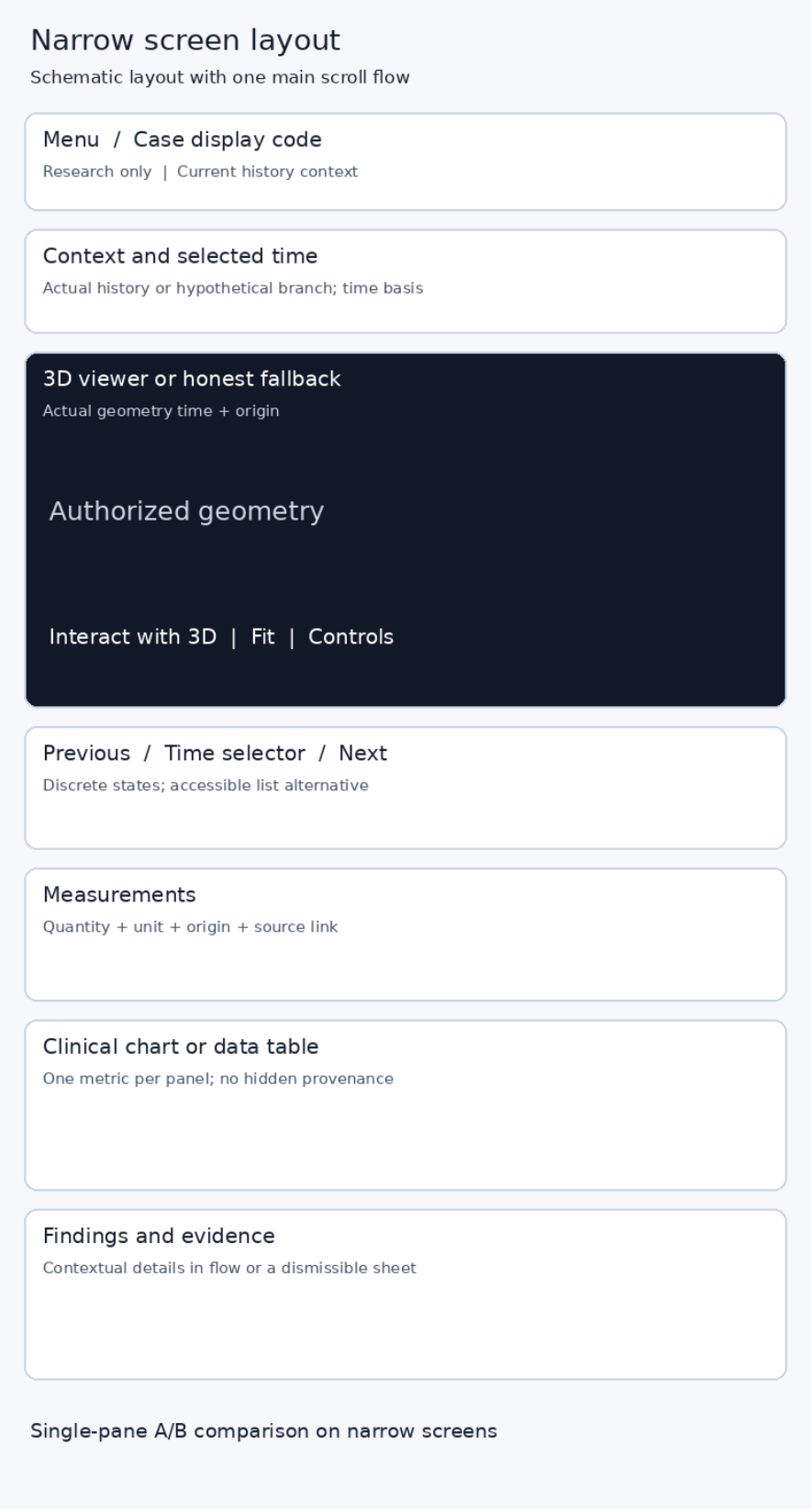
At desktop widths of at least 1200 CSS pixels, use a 200–224 pixel navigation rail, a flexible main column and a 320–380 pixel contextual inspector. Within the main column, place the context selector and selected-time summary above the viewer; place the event timeline directly below it; place chart panels below the timeline. The inspector uses Findings, Records and Evidence tabs. Keep clinical content in the main scroll flow rather than forcing several independent scroll regions.

The context selector reads Actual history or Hypothetical scenario plus its family and run label. The time heading shows the snapshot anchor. A separate image-time line identifies the geometry's actual source. The viewer toolbar exposes Rotate, Pan, Zoom, Fit view, Layers and Controls help. Display selected lesion name and validated measurement beside the viewer, with source and method one click away.

At 768–1199 pixels, collapse the navigation rail and place the inspector below the viewer or in a dismissible sheet. Below 768 pixels, stack context, time summary, viewer, timeline controls, measurements, charts and findings. Display one comparison pane at a time with a labeled A/B switch. Do not shrink chart text or hide provenance badges to retain a desktop arrangement. A 320 CSS pixel viewport must permit reading and non-3D workflows without page-level horizontal scrolling.

## 4.4 Layout references






These layouts define content placement and hierarchy. They do not prescribe clinical values or replace interaction requirements.

### 4.5 Scenarios workspace

Start with a branch library showing family, baseline, run version, publication status and date generated. Five family cards have short plain-language descriptions. Call SC-02 Modest change or stability rather than implying a formal clinical partial response classification. A read-only viewer opens published runs; editors can Generate scenario from the approved baseline.

The generation workflow has four steps: Choose baseline, Choose scenario and hypothetical exposure, Review assumptions, Generate and validate. Show required known clinical information and missing fields before submission. Unknown receptor status is never preselected positive. An ineligible option remains explainable with a reason; the interface cannot override constrained eligibility by hiding a warning. P0 does not expose unconstrained research mode.

The review screen lists timing, tumor pattern, clinical-condition dimension, seed and configured engine version. Separate scientific settings from advanced reproducibility details. No dose entry, treatment recommendation or best-treatment ranking exists. Generate starts a durable job; successful execution leads to validation review. Publish scenario is a distinct authorized action and appears only when the selected run is ready. Existing published runs remain usable while another run processes.

## 4.6 Add observation and correction

An editor opens Add observation from the case. Step one chooses a supported kind; step two records source evidence and collection time; step three enters code, value, unit and applicable context; step four reviews the complete submission. Date, timestamp, relative day and unknown time are distinct inputs. Unknown time may be stored where allowed but cannot be invented into a dated timeline state.

Units are chosen from the concept dictionary; abnormal values may be valid observations. Missingness uses a labeled reason rather than a blank assumed normal. Blood pressure has separately coded systolic and diastolic components unless a supported composite contract is introduced. Notes reference verified private documents. The form never lets an editor declare an unsupported entry authentic by selecting a provenance badge.

Submission is atomic. A 201 receipt means the observation was saved, not that a new mesh or state is ready. Show Saved — updating the research view, then offer Open new snapshot when publication succeeds. Keep the current snapshot stable until the user chooses to switch. Correction creates a new source-supported revision with a reason; the review shows original and replacement. Retraction has a separate confirmation naming its effect on the current view while retaining historical records.

## 4.7 Sources and evidence inspection

The case Sources page shows collection/release, rights summary, approved source identifiers, processing versions, annotation class and lineage. Protected mapping keys and storage paths are excluded. Clicking a measurement opens a record panel with exact value/unit, actual time, origin, method, source locator summary, quality and related artifacts. A visible label explains whether QC is automated, manually reviewed or pending.

An evidence citation opens the exact approved passage with document title, version/date, locator and canonical publisher link. Literature evidence is labeled Background evidence; patient records are labeled Case evidence. A passage cannot be displayed as proof of an unrecorded patient characteristic. Restricted text is summarized only within permitted rights; show Metadata only when full passage access is unavailable.

## 4.8 Activity and data administration

Activity presents queued, running, validating, succeeded, failed and cancelled work with visible results and safe retry guidance. A cancel request reads Cancellation requested until the server acknowledges a terminal state. Unknown progress shows an indeterminate indicator rather than an invented percentage. Reopening the job by ID restores server status.

The administrator import screen has staged files/manifest, rights review, identity linkage, imaging/QC checks and a publication summary. Invalid rows and failed checks link to actionable details. No branch or clinical record becomes viewer-visible through a green upload-complete indicator alone. Administrative source mapping and configuration remain separate from the judge-facing workspace.

## 4.9 Evaluation screen

Show only persisted reports with task, model/pipeline version, cohort definition, independent original patient count, split method, measurement definition and metrics. Label whether outcomes are real or synthetic. Report a simulation reproducibility check separately from model performance on actual observations. Missing evaluation reads Not evaluated; never display a decorative accuracy percentage.

For the hackathon, a compact report may include physical geometry checks, source linkage checks, scenario repeatability, grounding tests and usability observations. If no genuine repeated-imaging case is verified, show that release requirement as unmet. Evaluation publication and retrieval require the API amendment in section 17 or an authorized immutable report artifact already delivered through the artifact contract.

# 5 Provenance and status language

| Persisted provenance | Visible label | Marker and explanation |
|---|---|---|
| **REAL_OBSERVED** | Observed | Filled circle; source-recorded finding |
| **REAL_DERIVED** | Derived from observed data | Square; deterministic or documented processing of real input |
| **SYNTHETIC_GENERATED** | Simulated | Diamond and dashed outline; hypothetical generated value or geometry |
| **INTERPOLATED** | Interpolated estimate | Hollow circle; intermediate estimate, P1 only |
| **MODEL_PREDICTED** | Model estimate | Triangle; model-generated future estimate, disabled in P0 |

History context, item provenance, availability, QC and publication are separate concepts. In a hypothetical branch the shared baseline still reads Observed baseline, while later geometry reads Simulated follow-up. A measured source result inside an observed snapshot may have derived measurement provenance. An available item can still have a QC warning; a missing item is not automatically rejected.

Display Ready, Processing, Not recorded, Invalid data or Access restricted according to actual metadata and permission-safe messages. QC passed expands to Named technical checks passed; it must not expand to Clinically validated. Generated interpretations show AI-assisted summary, Previously generated summary or Rule-based summary according to actual execution mode. Do not label all saved results as live AI.

# 6 Visual design system

Use a restrained light research workspace with a dark neutral 3D viewport. The visual hierarchy prioritizes case identity, context, selected time, measurement and evidence. Avoid decorative whole-body silhouettes, glowing biological particles and invented physiological dashboards. The lesion is the primary visual asset; surrounding tissue is subordinate and appropriately labeled.

| Token | Proposed value | Application |
|---|---|---|
| **canvas** | #F6F8FB | Main background |
| **surface** | #FFFFFF | Cards, forms and inspector |
| **text_primary** | #172033 | Main body and headings |
| **text_secondary** | #475569 | Supporting labels |
| **border** | #CBD5E1 | Separation; not the sole focus/selection signal |
| **action** | #1D4ED8 | Primary buttons, links, selected controls |
| **observed** | #0F766E | Observed labels and series with circle markers |
| **derived** | #1D4ED8 | Derived labels and square markers |
| **simulated** | #6D28D9 | Simulated labels and diamond markers |
| **warning** | #92400E | Warning text with pale amber background |
| **error** | #B91C1C | Invalid or failed operation, with text explanation |
| **viewport** | #111827 | 3D background |
| **lesion** | #E9A38A | Neutral lesion material; never a severity heat map |

Token values are proposals; verify all actual text/background and chart combinations before claiming contrast compliance. Context and status use text plus shape or line style, not color alone. Red is not used as a generic tumor severity scale. A green indicator means an operation/check succeeded, not that a patient is healthy.

Use Inter where available with system sans-serif fallback. Body text defaults to 16 pixels with 1.5 line height; labels may be 14 pixels; page heading 28–32 pixels; section heading 20–24 pixels. Quantities use tabular numerals. Display precision follows measurement support, typically two or three significant digits, with full exact value in the record panel. Round only presentation; never recompute scientific comparisons from rounded strings.

Use an 8 pixel spacing scale with 4 pixel fine spacing. Cards use 12 pixel radius, a quiet border and minimal shadow. Prefer one primary action per task region. Buttons should normally be 40–44 pixels high. Icon-only controls require an accessible name and visible tooltip; high-stakes context changes use explicit text. Reduced motion disables animated transitions and auto camera movement.

# 7 Three dimensional interactions

## 7.1 Geometry and camera behavior

The client renders authorized precomputed GLB/glTF geometry using Three.js or React Three Fiber. The server remains authoritative for physical measurements. Preserve frame orientation, handedness and unit transformations from the manifest. The browser shall not normalize each timepoint to an identical displayed size: that would hide growth. Fit view establishes a consistent camera scale for a comparable series, with explicit refit available.

On first load, fit the verified bounds with margin and orient using frame metadata. Include an orientation indicator only when validated mapping is available; otherwise label Orientation unavailable rather than inventing anatomical directions. An arbitrary centered lesion is described as Lesion view, not as a registered breast position. A perspective scene does not support a simple screen-pixel ruler; omit a physical scale bar unless the chosen projection and scale calculation are verified. Numeric physical measurements remain visible.

Selecting another state of the same comparable lesion preserves camera orientation and scale when bounds remain sensible. Switching patients resets camera and selected layers. A major bounds change may offer Fit updated geometry instead of silently re-centering. Clipping prevention and pan limits must not crop a large lesion without notice. No auto-rotation runs on entry.

## 7.2 Control contract

| Action | Pointer or touch | Accessible alternative |
|---|---|---|
| **Rotate** | Drag in Rotate mode; one-finger drag inside active viewport | Labeled rotation buttons and front/side/top presets when frame-valid |
| **Pan** | Drag in Pan mode; two-finger drag where supported | Up, down, left and right pan buttons |
| **Zoom** | Wheel or pinch while interacting with viewport | Zoom in and Zoom out buttons |
| **Fit or reset** | Fit view or Reset view button | Same keyboard-focusable buttons |
| **Select lesion** | Click/tap rendered lesion | Lesion selector/list with name and source metadata |
| **Show layers** | Layers menu | Checkboxes for lesion and permitted context layers |
| **Leave viewport interaction** | Click outside or Escape | Tab moves to next control; no keyboard trap |

Canvas gestures must not consume scrolling for the entire page. On touch, require an explicit Interact with 3D action when needed to prevent accidental gesture capture, with Done interacting available. Do not make drag, pinch, wheel or right-click the sole way to complete any task. Show concise controls help on demand, not permanently over the lesion.

### 7.3 Layers and comparison

P0 layers include selected lesion and an optional verified or illustrative breast-region context. Labels identify layer origin. Illustrative context is off by default if its placement could imply unsupported anatomy. P1 may add adjustable opacity, mask outlines and wireframe. An opacity slider changes appearance only and cannot alter measurements.

Comparison defaults to two separate states with synchronized numeric summaries. Overlay is enabled only when frames and registration QC permit it. Label each state with time, provenance and marker. Equal camera framing does not imply registration. A split view can compare unregistered lesions but must state Spatial alignment not established. Multi-lesion cases require an explicit lesion choice; do not automatically sum unrelated measurements or assume a biological match.

### 7.4 Missing geometry and failure

If geometry is absent, show the valid measurements and No supported 3D geometry for this state. A last available scan may be shown only through a labeled historical-context option with its actual time; do not quietly fill the selected state. If a lesion has been resected in a configured scenario, show the intervention context and documented residual/absence semantics rather than an unexplained empty canvas.

WebGL failure, lost context or memory pressure switches to approved static views if available and a measurement table. Static previews must be derived from the same authorized geometry and carry its origin/time. If no preview exists, display metadata without a placeholder tumor. Provide Retry 3D and continue navigation. This fallback preserves information but does not count as passing the interactive 3D release test.

## 8 Timeline behavior and synchronization

The main timeline represents snapshot anchors in one explicit history context. The observed history and each hypothetical run have separate rails. Branch position zero references the shared real baseline; follow-ups use simulation-relative time. For real history, respect source dates, shifted dates or relative days and label the time basis. Unknown-time records appear in an Unplaced records section, not at an arbitrary point on the axis.

Use true elapsed spacing for compatible dated observations. When points crowd together, use zoom, grouping or a selectable list. If an ordinal stepper is used for easy selection, label it Observation sequence and show each actual time; never imply uniform intervals. The timeline has Previous, Next and a select-list alternative. A continuous drag preview may highlight a target marker but only existing discrete states can be committed in P0. No mesh morphing between visits.

The selected-state heading shows date/day, context and availability summary. Separate tracks show imaging examinations, laboratory collections, source notes and treatment events, using actual collection/event times. These tracks are not new snapshot records unless built and published through the API. A lab point without a matching snapshot opens its record; it does not silently change the entire workspace to a fabricated state.

On a new selection, establish a selection key containing project, patient, branch/run, state and lesion. Fetch and validate that snapshot. Ignore or abort late responses for earlier keys. Update the header, values, chart cursor, interpretation context and geometry manifest as one coherent selection. If geometry bytes are still loading, show a loading panel for that state rather than the prior mesh under the new date. Old results may remain only in an explicitly labeled Previous selection panel outside the active state.

Playback is optional P1 and steps between existing states, pauses on load/failure and announces discrete selection. If presentation pacing is uniform while observation intervals differ, label it presentation playback. Reduced-motion users retain manual navigation. Changing branch stops playback and resolves a supported baseline or chosen state; never infer that Day 60 exists in every branch.

## 9 Charts and quantitative presentation

### 9.1 Shared chart rules

Every chart includes metric name, unit, context, actual time basis, legend and a table-view toggle. Tooltips are available on focus, click and touch, not hover alone. They show exact time, displayed value, origin, source link and relevant QC. Numeric

observations use true elapsed spacing when compatible. Unknown time is excluded from the time plot with an explicit unplotted count and remains in the table.

No smoothing, spline interpolation or gap filling is enabled by default. A line connecting two measurements is labeled a visual guide, not evidence of continuously measured behavior. Simulated series use dashed lines and diamond markers; observed/derived points retain their distinct markers. A future interpolated series would require its own visible estimate label and cannot masquerade as recorded points.

Missing and invalid records do not plot at zero. Censored values such as below detection use comparator labels and threshold markers. Do not align incompatible source time domains. Tables preserve precise decimal strings; browser floating-point conversion is permitted for plotting only after bounds checks and cannot produce authoritative change metrics.

### 9.2 Required chart types

| View | P0 presentation | Rules |
|---|---|---|
| **Lesion burden** | Line/point chart of one selected volume or diameter metric | Keep volume and diameter separate; show units and annotation meaning |
| **Change comparison** | Two-state values, absolute change and percentage change from server | Zero baseline or incompatibility shows Unavailable with reason, never an invented percentage |
| **Laboratory trends** | Small multiples, one analyte/unit per panel | Reference bands only from applicable recorded intervals; no universal assumed normal band |
| **Vital signs** | Separate metric panels; linked systolic/diastolic components when supported | No mixed-unit dual axes; show component labels |
| **Symptoms** | Discrete observations or steps only when persistence is explicitly supported | Preserve severity scale; do not treat a 0–10 score as an oncology toxicity grade |
| **Treatment context** | Event markers and documented exposure intervals | Mark recorded versus hypothetical; unknown ends are visibly open, not invented |
| **Modality availability** | Labeled status grid per state | Availability is not health status and not a diagnostic heat map |

For interval bands that vary by laboratory context, draw only the applicable interval at the associated observation or clearly label intervals as variable. P0 may show interval values in tooltips/table instead of a continuous band. A patient's missing interval stays missing. Abnormal but valid results remain visible.

Within a two-state or branch comparison, use the same y-axis scale for equivalent metrics. If a truncated axis is useful for a single trend, label its bounds clearly; bar charts start at zero unless a justified alternative is clearly shown. Do not manipulate axes to suggest stronger response. Multi-branch charts are P1 and display at most three selected branches by default to preserve readability.

### 9.3 Chart and timeline linking

Selecting a published state sets a vertical cursor at its anchor and highlights current versus historical-context observations without inventing coincident collection times. Clicking a plotted source point opens the record and offers Open associated snapshot only when one exists. Pin A and Pin B compares explicit state IDs; the UI shows baseline/source identity and API comparability reasons. Invalid comparison never removes access to either state's valid underlying data.

## 10 Findings and evidence linked interpretation

The Findings panel begins with scope: selected case, state/time and observed or hypothetical context. Sections are Supported findings, Changes, Missing information and Limitations. Each factual finding links to case measurements or source observations; background reasoning links separately to literature passages. Do not display a medical confidence percentage unless a documented evaluated output supports its exact meaning.

Editors can request Generate research summary after viewing available inputs. If external processing is denied, explain that it is unavailable for this case and allow an approved deterministic summary when implemented. A provider outage leaves charts and source records usable. If a grounding check fails, show Summary could not be verified and a safe retry/action; do not show unvalidated partial model prose in the public panel.

Show execution mode, generation time and model/template details under About this summary. Saved interpretation is labeled Previously generated. A rule-based fallback is labeled Rule-based summary. If new records exist, show New information is available; this summary describes the selected earlier snapshot, with Open latest view. Do not mark a historical snapshot intrinsically wrong merely because newer data exists.

Evidence inspection preserves back navigation and selected state. A publication citation by itself is not proof of a case-specific conclusion. The UI must not use phrases such as Best treatment, Confirmed resistance, Survival probability or Predict outcome in P0 action labels.

# 11 Client state and interaction contracts

The URL owns the navigable selection; server responses own clinical data, publication and job state; component state owns camera, temporary form input and expanded panels. Use typed generated API contracts, keyed requests and bounded in-memory caches. Clear patient-bearing caches on logout or authorization change. Do not put patient data into persistent browser storage, service-worker offline caches or analytics payloads by default.

| User action | Required behavior |
|---|---|
| **Select a different patient** | Clear prior clinical panels immediately; fetch new authorized case; reset camera |
| **Select another state quickly** | Cancel/ignore stale requests using complete selection key; no old/new panel mixture |
| **Open a saved link** | Authorize all IDs; restore exact selection or explain inaccessible/withdrawn status |
| **Submit observation or simulation** | Disable duplicate submit while pending; retain idempotency key for the same deliberate attempt |
| **Receive 412 conflict** | Preserve unsent form; refresh version and show changes; never automatically overwrite |
| **Receive 422 validation** | Field-level messages, focusable error summary and preserved safe input |
| **Receive 202 job receipt** | Show accepted/queued status and poll authorized job; never announce completion |
| **Receive 429** | Honor Retry-After; explain wait without duplicating job |
| **Asset URL expires** | Request one new authorized grant; do not reuse indefinitely or loop on forbidden access |
| **Resource withdrawn** | Remove it from active display, explain status, preserve safe navigation |

Job polling starts with server Retry-After, slows with backoff and pauses when the tab is hidden where appropriate. On returning, refresh server status. Cancellation changes local copy only after the server response. Leaving a screen does not cancel a durable job. A lost response requires checking/replaying the same idempotent command, not a new seed or key that creates another run.

# 12 Loading empty error and access states

Use region-level skeletons for metadata and a real Loading geometry message for the viewer. Loading artwork must not resemble a patient lesion. Error messages state what failed, what remains available and one useful next action. No technical stack traces or source keys appear in user messages.

| Condition | Required message and recovery |
|---|---|
| **No source laboratories** | No laboratory results were recorded for this case. Explore a labeled scenario if available. |

Toasts are supplementary; durable outcomes appear in the relevant panel or activity record. Destructive administrative withdrawal and retraction require a concrete confirmation with resource identity and reason. Ordinary date selection, camera actions and read-only exploration require no confirmation.

# 13 Accessibility and responsive acceptance

Target WCAG 2.2 AA for the interface and provide a meaningful non-canvas route to all research information [R1]. This is a target requiring testing, not an established conformance claim. The numerical requirements below are project acceptance rules selected to support that target, not a paraphrase of every WCAG criterion.

Normal text contrast must reach 4.5:1; large text 3:1; essential control boundaries, focus and chart markers 3:1 against adjacent colors where applicable. Controls have clearly visible focus and are not obscured by sticky elements. Use 44 by 44 CSS pixel targets where practical, at least 24 by 24 or valid WCAG spacing exceptions for smaller controls. Do not rely on color, hover, drag or complex gestures alone.

Provide semantic headings, landmarks, labels and a Skip to content link. Forms have explicit labels, instructions and grouped fields. Error summaries link to affected fields. Dialogs trap focus only while open, close with Escape when safe and return focus to the trigger. Charts provide equivalent tables and concise narrative summaries. Canvas has a descriptive name, an accessible lesion/measurement summary and button-based camera controls.

Timeline slider behavior follows the APG pattern with arrow keys, Home/End and an understandable value text such as Simulated day 60, state 3 of 4 [R2]. The equivalent select list and Previous/Next controls remain available, especially for touch assistive technology. Tabs use correct roles and manual activation when content requires loading [R3]. Navigation links are not incorrectly modeled as in-page tabs.

Use polite live announcements for selected time and completed updates; reserve urgent alerts for actionable failures. Do not announce every camera frame or polling tick. Honor reduced motion, 200 percent zoom, browser text resizing and reflow at narrow widths. Test keyboard-only use, screen reader reading order, touch, color-vision differences and low-powered graphics. All essential data remains available without WebGL; full 3D interaction still requires a separate supported-device test.

# 14 Performance and resource budgets

Use prepared meshes and approved compact previews for the demo. Lazy-load the 3D library and optional slice viewer. Load one patient and active mesh by default; prefetch at most the adjacent authorized state on adequate connections. Dispose obsolete GPU buffers, textures, render targets and event listeners. Never download a complete MRI collection or segmentation checkpoint into the browser.

| Metric | Proposed target and measurement |
|---|---|
| **Warm metadata response** | Architecture target p95 below 700 ms; measure deployed API separately from full page load |
| **State manifest** | Architecture target p95 below 1.5 seconds |
| **First usable prepared mesh** | Within 5 seconds on the declared test connection/device; record cold and warm results |
| **Camera interaction** | Aim for at least 30 frames/second on the declared Ryzen 5 3500U/8 GB test device; benchmark, do not assume |
| **Active mesh transfer** | Start with a target of at most 5 MiB per demo GLB; measured exceptions or an optimized derivative required |
| **UI response to selection** | Immediate loading/selection feedback, with no fabricated completed state |

These are engineering targets, not measured claims. The actual laptop graphics capability must be tested rather than inferred from a memory figure. Cap rendering resolution on high-density screens, reduce context geometry, pause

rendering when idle/hidden and offer a low-detail mode. Optimized visual meshes retain parent provenance and cannot silently redefine validated mask volume. A heavy 2D MRI viewer remains optional if it prevents reliable P0 execution.

# 15 Interactive demonstration specification

## 15.1 Readiness and modes

Prepare at least five approved curated case entries where feasible, with one hero baseline and all five published scenario families, plus at least one separately verified genuine repeated-imaging example. A case may serve more than one purpose; counts must reflect distinct real patients. Each demo entry pins authorized patient, state, branch/run, geometry and interpretation IDs and a readiness status. Nothing in the presentation mode grants additional data access.

Provide two explicit modes: Explore approved demo for read-only judges, and Presenter workspace for authorized live writes. A presenter uses an isolated demo project with source-supported or clearly simulated content. Guided steps navigate actual application routes and controls; they do not replace them with prerecorded animations presented as a live system.

A guided sequence configuration may be bundled as non-sensitive IDs and titles while every underlying request remains authorized. No signed URLs, protected source keys or credentials are embedded. The sequence verifies required published resources before enabling Start. If a step is unavailable, show it as unavailable and explain the dependency instead of substituting a different case invisibly.

## 15.2 Five minute guided sequence

| Time budget | Step and interaction | Proof visible to the audience |
|---|---|---|
| 0:00–0:30 | Open approved hero case and source summary | Authentic baseline, available modalities and research scope |
| 0:30–1:15 | Rotate lesion, inspect measurement and annotation source | Physical measurement, derived geometry and traceable mask meaning |
| 1:15–2:10 | Open a published hypothetical branch and choose three follow-ups | Shared observed baseline; simulated geometry and charts with discrete dates |
| 2:10–2:40 | Switch to SC-05 and inspect clinical/tumor trends | Clinical deterioration need not mirror lesion burden; no causal claim |
| 2:40–3:25 | Open genuine repeated-imaging case and compare two verified times | Actual longitudinal changes, matched meaning and source evidence |
| 3:25–4:10 | Presenter submits a prepared source-backed observation, or shows its prior acknowledged update | New revision and processing status; earlier state unchanged; no fake new MRI |
| 4:10–4:45 | Open grounded summary and follow one source/evidence link | Source-supported finding, missingness and execution-mode disclosure |
| 4:45–5:00 | Open evaluation/limitations summary | Measured test evidence and clearly unmet or deferred capabilities |

Timing is rehearsal guidance, not a processing-speed guarantee. If a live job exceeds the available presentation time, show its actual queued/running status and offer a separately labeled previously completed result. Never rename precomputed output as the result of the pending command. A judge may leave the guide at any step and continue normal exploration.

## 15.3 Demo controls and recovery

Controls are Previous step, Next step, Restart guide and Exit guide. A progress indicator reads step N of M with a short title. Restart guide resets selection, camera and guide progress only; it does not delete observations, cancel jobs, regenerate branches or overwrite source data. Clearing demo writes is a separate administrator operation outside the public interface.

Preflight checks authorization, published manifests, permitted artifact grants, supported WebGL, prepared summaries and genuine follow-up availability. The live provider status does not block approved historical summaries. A network interruption shows the current limitation. A recorded demonstration may be supplied separately and clearly identified; it is not an offline copy of protected patient data or a disguised live session.

Do not make a fresh segmentation model run, large upload or uncertain inference entitlement a required step in the five-minute sequence. The scientific requirement for a genuine longitudinal case cannot be satisfied by a fallback animation or synthetic branch.

# 16 Component and interaction deliverables

Implement reusable CaseHeader, ContextSelector, ProvenanceBadge, TimeLabel, ModalityStatus, MeasurementCard, ViewerToolbar, LesionViewer, TimelineSelector, ClinicalChart, DataTable, RecordInspector, EvidenceDrawer, JobStatusPanel, ErrorSummary and DemoStepper components. Each has loading, empty, invalid, forbidden and success states where applicable. Typed props carry IDs, origin, time and quality instead of inferring them from colors or labels.

Required design deliverables are desktop and narrow-screen wireframes, token definitions, component state examples, keyboard behavior notes, route map and demonstration script. Required implementation evidence is a working vertical slice that opens one verified case, rotates its actual derived mesh, changes snapshot, reads a chart/table and resolves one source reference. Static mockups alone do not satisfy interactive readiness.

# 17 API integration and explicit contract amendments

Use the project-scoped /api/v1/projects/{project_id} base from COS-SPEC-008. Never substitute earlier preliminary imaging paths from COS-SPEC-005 or the architecture's illustrative envelope. The following mappings describe consumption of existing contract definitions, not proof of implemented services.

| Screen or action | Existing contract dependency |
|---|---|
| **Case library and overview** | GET /patients and /patients/{p}; use only supported filters |
| **Twin selection and charts** | GET /patients/{p}/states and /states/{s}, /observations, /studies and study detail; fetch every needed cursor page |
| **Source/geometry inspection** | GET /artifacts/{a}, /segmentations/{seg}, /resources/{id}/provenance; POST artifact read-grants |
| **Two-state comparison** | GET /patients/{p}/comparisons with explicit metric and lesion IDs |
| **Scenario generation** | GET /scenario-definitions; POST eligibility-checks, branches, branch runs and publications |
| **Add or correct observations** | POST observation-batches, corrections or retractions with required preconditions |
| **Summary generation and evidence** | POST interpretation-jobs; GET interpretations/{i} and evidence passages |
| **Import and validation** | Upload-session completion, imports, import publications and resource validations |
| **Processing status** | GET /jobs/{j}; authorized cancellation command |

COS-SPEC-008 lacks complete discovery/read contracts needed by some screens. The following additions are proposed amendments, not silently existing endpoints. They require schema review, access tests and inclusion in generated clients before their UI controls are enabled.

| Proposed amendment | Minimum shape and safeguards |
|---|---|
| **GET /capabilities** | enabled feature codes, permitted actions, supported dictionaries/units and limits; project/role scoped; no secrets or inferred clinical eligibility |
| **GET /patients/{p}/branches** | Published branch list for viewers; authorized draft/run metadata for editors; cursor envelope and baseline references |
| **GET /patients/{p}/observations/{o}** | Authorized exact revision plus safe source/time/subtype metadata; needed when summary lists do not embed full detail |
| **GET /patients/{p}/interpretations** | Discover summaries by state_id and snapshot hash, execution mode, freshness and publication; do not guess interpretation IDs |
| **GET /jobs** | Authorized project/patient/status filters and cursor pagination for Activity; otherwise show only known current-session job IDs |

| Proposed amendment | Minimum shape and safeguards |
|---|---|
| GET /evaluations and /evaluations/{id} | Approved versioned reports, task/cohort, real versus simulated outcomes, methodology, metrics, limitations and artifact references |
| Chart query semantics | Define time_domain, context, pagination and numeric/censored variants explicitly for observation histories; do not plot only the first page as complete history |

Until capability discovery is implemented, a server-controlled, permission-checked feature configuration can supply enabled controls. Until branch/summary discovery exists, the curated demonstration may pin approved IDs in its sequence manifest. This is a bounded demo workaround, not a scalable patient dashboard contract. Dictionary options must come from reviewed server configuration; the browser cannot invent unit conversions or eligibility rules.

The browser and API must agree on all chart/member references. StateDetail membership, actual observation times, temporal_relation and measurement metadata must be available either inline or through authorized detail reads. If they are not, that is an integration blocker, not an invitation to infer missing fields. No existing source specification is modified by this document; amendments require a new COS-SPEC-008 revision during implementation.

# 18 Acceptance and usability evaluation

| ID | Scenario | Pass condition |
|---|---|---|
| UX-01 | Open and reload a case link | Same authorized patient/context/snapshot; no data from another case |
| UX-02 | Rotate, pan, zoom and fit geometry | Pointer, touch and button alternatives work; measurements do not change |
| UX-03 | Select a larger comparable lesion state | Consistent physical scale and camera; growth not hidden by independent auto-fit |
| UX-04 | Change timeline rapidly under delayed responses | Header, chart cursor, values and mesh belong to the same selected state; stale responses ignored |
| UX-05 | Inspect a lab-only update | No new imaging is implied; historical geometry is optional and visibly dated |
| UX-06 | Navigate a hypothetical branch | Observed baseline and at least three simulated follow-ups are distinguishable by text and marker |
| UX-07 | Inspect MRI phases | Phase change remains inside one examination and creates no visit |
| UX-08 | Missing, unknown-time or censored data | Appropriate empty/censored presentation; never zero or invented timestamp |
| UX-09 | Compare incompatible measurements | Clear unavailable result with reason; no calculated percentage |
| UX-10 | Open a case measurement or claim | Correct authorized exact record/passage; back returns to the selected state |
| UX-11 | Save an observation or handle conflict | Durable receipt/job shown; preserved history; 412/422 does not lose safe input or overwrite data |
| UX-12 | Run and publish a scenario | Duplicate prevention, eligibility explanation and no partial branch visible |
| UX-13 | Provider or WebGL failure | Honest state, usable remaining data, clearly labeled fallback |
| UX-14 | Keyboard and screen-reader walkthrough | Complete case/time/table/evidence tasks without canvas-only interactions or focus traps |
| UX-15 | Narrow screen and zoom | Reflow, readable units and origin labels; no hidden essential actions |
| UX-16 | Judge role attempts a write | No write action exposed; server also rejects unauthorized command |
| UX-17 | Demo preflight and guide reset | Missing dependencies visible; restart changes UI selection only |

| ID | Scenario | Pass condition |
|---|---|---|
| UX-18 | Evaluation screen | Report provenance and sample basis visible; no invented accuracy or independent synthetic patient count |

Run moderated task sessions with at least three participants when feasible: one researcher or clinician reviewer, one technically informed judge proxy and one user unfamiliar with the project. This is formative usability work, not clinical validation. Ask each to identify an observed versus simulated finding, navigate a timepoint, explain a missing chart, locate evidence and compare two states. Record task completion, errors, time and misunderstandings. Any confusion that turns a simulated value into a believed real observation is a release-blocking design defect.

Automated tests cover selection races, route restoration, role-specific controls, formatting, chart data membership and error states. Manual tests cover 3D scale, visual provenance, keyboard flow, touch and readability. Record actual performance measurements and tested device/browser versions; do not claim all-browser or accessibility conformance from a screenshot review.

# 19 Implementation sequence

1. Build tokens, shell, route selection, provenance/time labels and all empty/error states with clearly synthetic engineering fixtures.
2. Integrate one approved real case and artifact permissions. Verify lesion scale and actual source details before visual polish.
3. Implement snapshot selection, synchronized charts/tables, race handling and two-state comparison.
4. Integrate scenario eligibility, published branches, job states and the observation-update flow.
5. Add grounded findings, evidence inspection, report delivery and the explicit API amendments required by the selected screens.
6. Rehearse the guide, test permissions/failures/accessibility, measure performance and resolve blocking provenance or synchronization defects.

P1 adds optional slice review, registered overlay, side-by-side branch comparison, advanced camera controls and carefully labeled interpolation. These must not delay the functioning P0 research journey or weaken the source-versus-simulation distinction.

# 20 References and traceability

| Source document | Requirements carried into this specification |
|---|---|
| COS-PRD-001 Product Requirements | Case, twin, scenario, update and evaluation journeys; role boundaries; five families and genuine follow-up demonstration |
| COS-SPEC-003 Clinical Data and Dataset | Source/missingness fidelity, actual collection times, measurement units and protected identity |
| COS-SPEC-004 Synthetic Longitudinal Data | Shared authentic baseline, three or more generated follow-ups, reproducibility and exposure eligibility |
| COS-SPEC-005 Medical Imaging and Digital Twin | Physical scale, geometry layers, supported camera controls and dated historical imaging |
| COS-SPEC-006 AI and Multimodal Intelligence | Source-linked interpretation, uncertainty, evidence distinction and failure/abstention |
| COS-SPEC-007 System Architecture | Next.js/TypeScript, precomputed meshes, private grants, asynchronous work and constrained local hardware |
| COS-SPEC-008 Database and API Contracts | Canonical provenance, immutable snapshots, time domains, ownership, jobs, errors and concurrency |

The architecture and database/API documents were reviewed as current full-text sources; the PRD and imaging source sections were checked for screen, timeline and visualization requirements. Earlier detailed source requirements remain

subject to their stated verification gates. This UI specification does not certify dataset rights, model availability or clinical accuracy.

External accessibility references checked 22 October 2026. Accessibility standards support the interaction requirements; the application layout, palette, performance budgets and demo sequence are proposed CancerOS design decisions.

- R1 W3C Web Content Accessibility Guidelines 2.2: https://www.w3.org/TR/WCAG22/
- R2 W3C ARIA Authoring Practices Slider Pattern: https://www.w3.org/WAI/ARIA/apg/patterns/slider/
- R3 W3C ARIA Authoring Practices Tabs Pattern: https://www.w3.org/WAI/ARIA/apg/patterns/tabs/